Statistical Business Analysis Report

Project Overview

This project analyzed sales and customer churn data using descriptive statistics, visualizations, hypothesis tests, confidence intervals, and linear regression. The goal is to summarize patterns and identify possible business actions.

Data Summary

- Sales records analyzed: (100, 7)
- Customer records analyzed: (500, 9)
- Average total sales per record: 123650.48
- Average monthly charges: 113.636
- Customer churn rate: 10.60%

Key Findings

1. **Sales:** North region has the average sales of 142272.68 and the product (Laptop) average sales of 162050.42
2. **Distributions and correlations:** Tenure and churn had a moderate negative correlation of −0.51, meaning customers with longer tenure tended to be less likely to churn in this dataset.
3. **Customer churn:** Month-to-month customers had the highest churn rate at 20.59%, compared with 6.94% for two-year contracts and 4.30% for one-year contracts.

Hypothesis Tests

A p-value below 0.05 is commonly treated as statistically significant.

Test	Statistic	p-value	Conclusion
ANOVA: sales across regions	2.1644	0.0972	Significant if p < 0.05
Weekday vs. weekend sales	-0.4266	0.6721	Significant if p < 0.05
East vs. West sales	2.0623	0.0455	Significant if p < 0.05

Confidence Interval

The estimated average total sales per record are **123,650.48,** with a 95% confidence interval from **103,776.35** to **143,524.61**.

This interval estimates a range for the population mean; it does not contain 95% of individual sales records.

Regression

The regression examines the relationship between quantity and total sales.

- Slope: **26629.54**
- R-squared: **0.473**
- p-value: **0.0000**

In this dataset, the model accounts for approximately **47.3%** of the variation in total sales. This describes an association and does not prove that quantity alone causes sales to change.

Recommendation

The month-to-month group had a 20.59% churn rate, higher than the one-year and two-year groups. The business could investigate what is driving cancellations in this group and test retention offers. This pattern is an association, so it does not show that contract length causes churn.

Limitations

The churn dataset contains 500 customer records, and the sales dataset contains 100 records. These samples may not represent all customers or sales. The analysis uses the fields available in the CSVs, and some measures may be skewed. Statistical associations do not prove causation.

Conclusion

The analysis found that month-to-month customers had the highest observed churn rate at 20.59%. This suggests a useful area for further investigation, but more data and testing would be needed before making major business decisions.